Pouyan Navabi

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Research Interests

Physical Chemistry, Machine Learning, Atomic Layer Deposition, Thin Films, Electrochemistry

Education

University of Illinois Chicago — Chicago, IL
PhD in Physical Chemistry (GPA 3.67) — *Jan 2025 – Present*

University of Illinois Chicago — Chicago, IL
MSc in Chemistry (GPA 3.67) — *Aug 2022 – Dec 2024*

Sharif University of Technology — Tehran, Iran
MSc in Organic Chemistry (GPA 3.86) — *Sep 2019 – Feb 2022*

Isfahan University of Technology — Isfahan, Iran
BSc in Applied Chemistry (GPA 3.43) — *Sep 2015 – Feb 2019*

Research Experience & Publications

Machine Learning for Atomic Layer Deposition — 2025 – Present

- Developed physics-informed Bayesian active learning and generative ML models to predict ALD growth rates and optimize pulse times in real time from process parameters.
  - Shape-constrained Bayesian active learning of self-limiting saturation curves. *arXiv preprint* (arXiv:2606.26577) — 2026
  - Tuning of atomic layer deposition pulse time through physics-informed Bayesian active learning. *J. Vac. Sci. Technol. A (Editor's Pick)* (doi: 10.1116/6.0005430) — 2026
  - Advancing atomic layer deposition through artificial intelligence: Generative modeling and predictive analytics for TiO_2 growth rate prediction from TDMAT/O_3. *Surf. Coat. Technol.* (doi: 10.1016/j.surfcoat.2026.133341) — 2026
  - Artificial intelligence in atomic layer deposition. *J. Vac. Sci. Technol. A (Featured)* (doi: 10.1116/6.0004881) — 2025

Atomic Layer Deposition on Organic Substrates — 2025 – Present

- Developed ALD processes on organic substrates and studied in-situ surface chemistry during early growth for biomedical and dental applications.
  - Color stability and surface roughness of printed provisional and definitive restorative materials functionalized with titanium dioxide nanocoating. *(Submitted)* — 2026

Interfacial Electrochemistry of Self-Assembled Monolayers — 2023 – 2024

- Modeled proton-coupled electron transfer in graphene nanoribbon SAMs through Python voltammogram simulation and experiment.
  - Interfacial electrochemistry of catalyst-coordinated graphene nanoribbons. *J. Am. Chem. Soc.* (doi: 10.1021/jacs.4c05250) — 2024

Material Synthesis and Characterization for Li–O_2 Batteries — 2023 – 2024

- Designed catalysts, electrolytes, and additives to improve the performance and stability of lithium–oxygen batteries.
  - Stabilizing lithium superoxide formation in lithium-air batteries by Janus chalcogenide catalysts. *Nano Energy* (doi: 10.1016/j.nanoen.2024.110510) — 2025

Technical Skills

Laboratory: ALD, Spectroscopic Ellipsometry, SEM, TEM, AFM, XRD, IR, UV-Vis, NMR, TGA, Optical Profilometry, Wet Lab (familiar with XPS, FIB-SEM, BET, HPLC, GC, Mass Spectrometry)
Programming: Python, LaTeX (familiar with MATLAB, Spartan, Gaussian)
Languages: Farsi (Native), English (Fluent)